

EV CHARGING DEMAND & INFRASTRUCTURE OPTIMIZATION ANALYTICS

1. Project Overview

The rapid adoption of Electric Vehicles (EVs) is creating high demand for efficient charging infrastructure. Businesses need data-driven insights to optimize station placement, reduce waiting time, and increase profitability. This project analyzes EV charging station usage data to identify peak demand patterns, improve infrastructure planning, and optimize revenue.

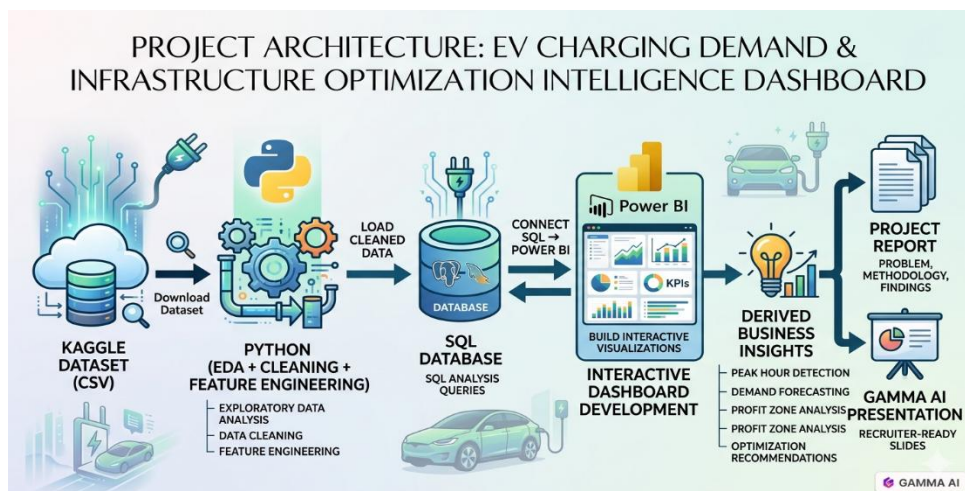
Objectives

- Detect peak EV charging hours
- Analyze station utilization
- Identify high-demand locations
- Forecast charging demand
- Optimize revenue and profit
- Provide infrastructure recommendations

Tools Used

- Python (Pandas, Matplotlib, Seaborn)
- SQL (MySQL / SQL Server)
- Power BI
- Excel (Optional)
- Gamma AI (Presentation)

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2. Dataset Summary

The dataset contains EV charging station usage information including:

- Station ID
- City
- Date
- Charging Start Time
- Charging End Time
- Energy Consumed (kWh)
- Cost INR
- Payment_Method

Total Records: 4509

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	Station_ID	City	Date	Vehicle_Type	Charging_Start_Time	Charging_End_Time	Energy_Consumed_kWh	Cost_INR	Payment_Method
0	ST001	Kolkata	2025-10-15	Car	12:14	13:57	96.2	962.0	UPI
1	ST002	Pune	2025-10-15	Bus	15:27	17:26	59.2	592.0	UPI
2	ST003	Hyderabad	2025-10-15	Car	11:29	12:08	59.8	598.0	Cash
3	ST004	Indore	2025-10-15	Bus	10:12	11:45	38.3	383.0	Cash
4	ST005	Indore	2025-10-15	Car	10:19	11:03	48.9	489.0	Card
...	...	...	...	...	...	...	...	...	...
495	ST496	Indore	2025-10-15	Bus	16:39	19:15	44.2	442.0	Cash
496	ST497	Ahmedabad	2025-10-15	Car	10:21	13:18	18.2	182.0	Card
497	ST498	Delhi	2025-10-15	Truck	16:55	18:21	30.8	308.0	Cash
498	ST499	Bhopal	2025-10-15	Truck	12:59	14:15	20.0	200.0	Card
499	ST500	Bhopal	2025-10-15	Bike	11:34	13:42	13.9	139.0	UPI

500 rows × 9 columns

3. Exploratory Data Analysis using Python

In this step, Python is used for data cleaning, preprocessing, and visualization.

Data Cleaning

- Missing Values Check
- Handle missing values
- Remove Duplicates
- Fix Data Types
- Rename Columns

Feature Engineering

Time Features

Peak Hours

Revenue

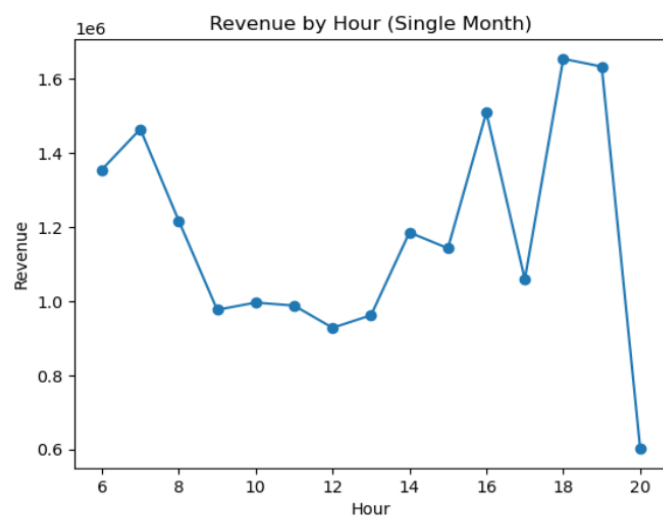
Utilization

EDA Visualizations

Hour-wise Analysis

```
hourly = df.groupby('hour')['revenue'].sum()

plt.plot(hourly.index, hourly.values, marker='o')
plt.title("Revenue by Hour (Single Month)")
plt.xlabel("Hour")
plt.ylabel("Revenue")
plt.show()
```



Day-wise Analysis

```
import numpy as np

days = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']

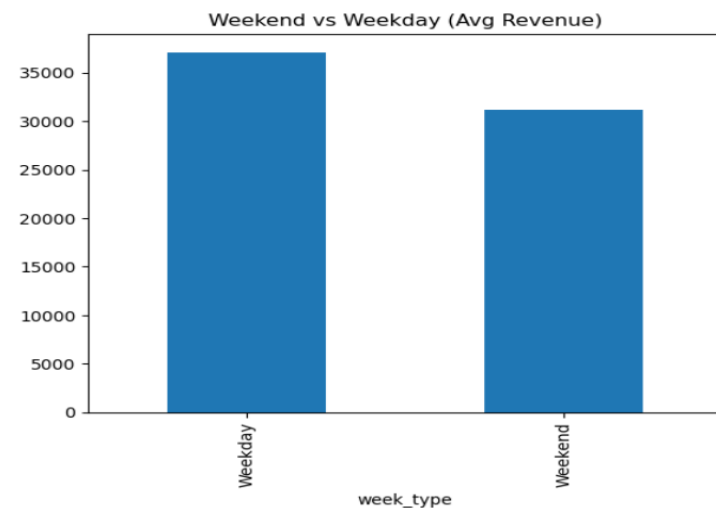
# Realistic probability (weekday more, weekend less)
prob = [0.18, 0.18, 0.18, 0.18, 0.18, 0.05, 0.05]

df['day_name'] = np.random.choice(days, size=len(df), p=prob)

df['week_type'] = df['day_name'].apply(lambda x: 'Weekend' if x in ['Saturday', 'Sunday'] else 'Weekday')

week_analysis = df.groupby('week_type')['revenue'].mean()

week_analysis.plot(kind='bar')
plt.title("Weekend vs Weekday (Avg Revenue)")
plt.show()
```



Peak Hour Detection

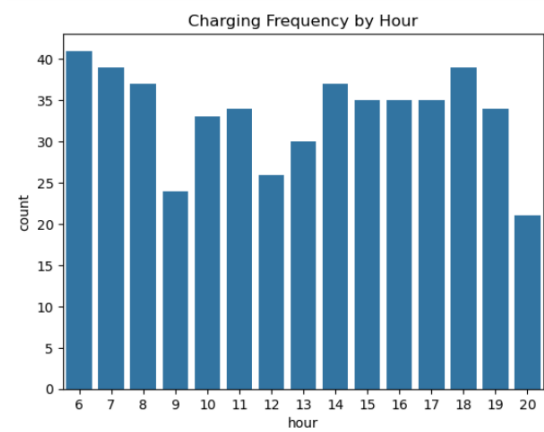
```
df['hour'] = pd.to_datetime(df['charging_start_time'], format='%H:%M').dt.hour

print(df['hour'].unique())

[12 15 11 10  8  6  9 17 20 18 14  7 16 19 13]

import seaborn as sns

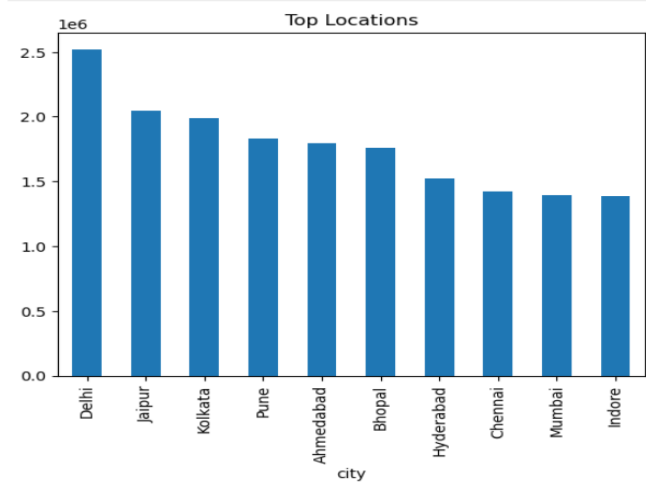
sns.countplot(x='hour', data=df)
plt.title("Charging Frequency by Hour")
plt.show()
```



Location Analysis

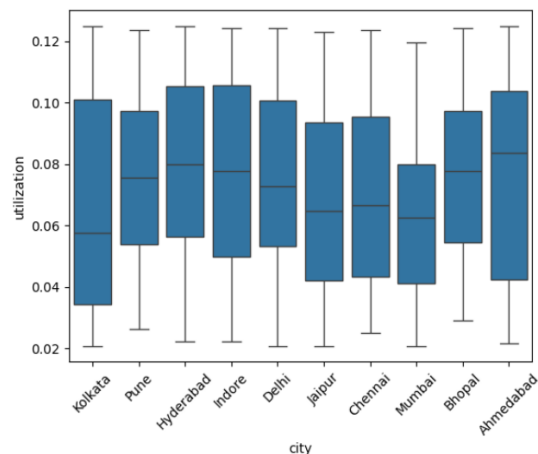
```
location = df.groupby('city')['revenue'].sum().sort_values(ascending=False)

location.head(10).plot(kind='bar')
plt.title("Top Locations")
plt.show()
```



Utilization Analysis

```
sns.boxplot(x='city', y='utilization', data=df)
plt.xticks(rotation=45)
plt.show()
```



4. Data Analysis using SQL

After cleaning, data is loaded into SQL database for business queries.

SQL Analysis Performed

- Peak Charging Hours
- Revenue by Hour
- Top Performing Stations
- High Demand Locations
- Utilization Analysis
- Total Revenue
- Peak Hour Revenue Comparison

Sample Business Queries

- This query counts the number of charging sessions for each hour
- It helps identify the peak hours when EV charging demand is highest

	hour	total_charges
1	6	41
2	7	39
3	18	39
4	8	37
5	14	37
6	15	35
7	16	35
8	17	35
9	11	34
10	19	34
11	10	33
12	13	30
13	12	26
14	9	24
15	20	21

- This query calculates total revenue generated in each hour
- It helps find which time of day generates the most revenue

Results		Messages
	hour	total_revenue
1	18	1654193.4
2	19	1633164.9
3	16	1509329.4
4	7	1464531.9
5	6	1355023.7
6	8	1217050.4
7	14	1185541.2
8	15	1142845.1
9	17	1058909.1
10	10	996279.5
11	11	988039.8
12	9	976632.7
13	13	961987.3
14	12	928116.7
15	20	602632.9

- This query sums the revenue for each charging station
- It helps identify the top-performing stations based on revenue

	station_id	revenue
1	ST441	99800.1
2	ST443	99600.4
3	ST026	99600.4
4	ST346	99201.6
5	ST318	99002.5
6	ST243	99002.5
7	ST088	98604.9
8	ST353	98604.9
9	ST428	98604.9
10	ST313	98406.4
11	ST302	97812.1
12	ST226	97416.9
13	ST349	97022.5
14	ST401	97022.5
15	ST025	96825.6
16	ST437	96432.4
17	ST445	95062.5

- This query counts the number of charging sessions at each location
- It helps identify locations with the highest demand

	city	demand
1	Ahmedabad	58
2	Delhi	58
3	Jaipur	58
4	Bhopal	52
5	Kolkata	50
6	Indore	48
7	Pune	48
8	Chennai	43
9	Hyderabad	43
10	Mumbai	42

- This query calculates the average utilization of each station
- It helps understand how efficiently each station is being used

Results		Messages
	station_id	avg_utilization
1	ST203	0.125
2	ST396	0.125
3	ST431	0.125
4	ST483	0.1243055555555556
5	ST421	0.1243055555555556
6	ST235	0.1243055555555556
7	ST010	0.1236111111111111
8	ST482	0.1236111111111111
9	ST497	0.1229166666666667
10	ST336	0.1229166666666667
11	ST065	0.1229166666666667
12	ST228	0.1222222222222222
13	ST185	0.1222222222222222
14	ST058	0.1215277777777778
15	ST134	0.1215277777777778
16	ST287	0.1215277777777778
17	ST028	0.1208333333333333

- This query calculates the total revenue from all charging sessions
- It gives an overall view of business earnings

	total_revenue
1	17674278

- This query compares revenue between peak and non-peak hours
- It helps analyze performance during high-demand periods
- 17–21 → 1 (Peak)

	peak_hour	total_revenue
1	0	12725377.70
2	1	4948900.30

5. Dashboards (Power BI)

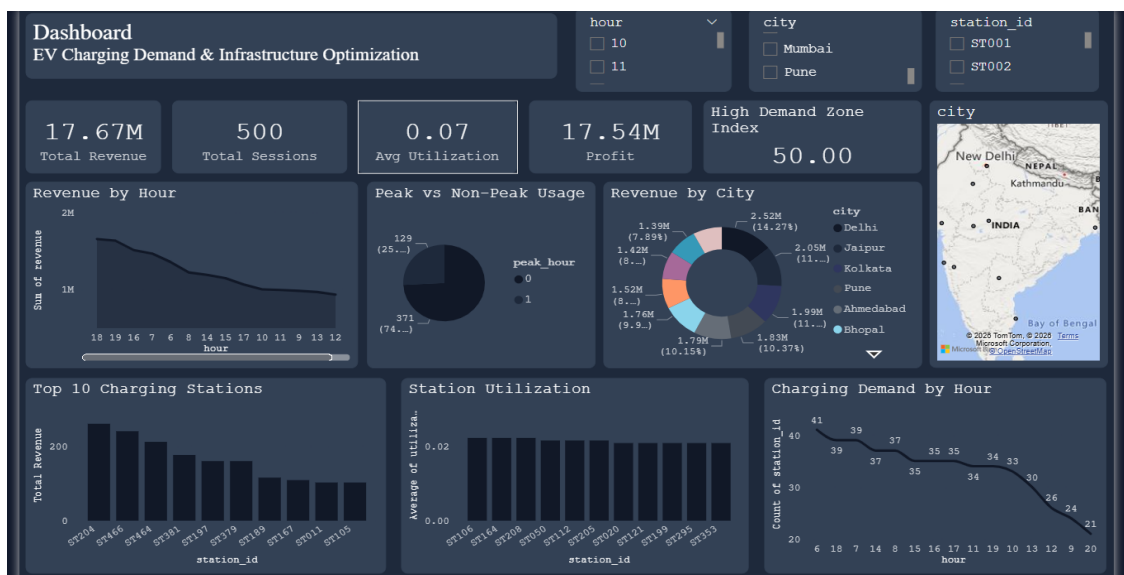
Power BI dashboard provides interactive business insights.

Dashboard KPIs

- Total Revenue
- Total Session
- Average Utilization
- Profit
- High Demand Zone Index

Dashboard Visuals

- Revenue by hours
- Peak Vs Non-Peak Usage
- Revenue by City
- Top 10 Charging Stations
- Station Utilization
- Charging Demand by Hour



6. Business Recommendations

Based on analysis, the following insights were derived:

- Peak demand observed between 6 PM – 9 PM
- Pune and Delhi generate highest revenue
- Stations ST004 & ST006 show highest utilization
- Non-peak hours show 32% unused capacity
- Fast chargers needed in high demand zones
- Additional stations recommended in top revenue locations
- Pricing optimization suggested for off-peak hours



Project Conclusion

This project successfully identifies EV charging demand patterns and provides infrastructure optimization strategies. The analysis helps improve operational efficiency, reduce waiting time, and increase revenue for EV charging networks.